

LangSlayer:

Slaying Hyper-Technical
Jargon into Simple Terms

CyberSlayerCorpz | Built for Suvidha AI Hackathon

Agenda

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Empowering ESL Students to Access Complex Text

LangSlayer is a fast, lightweight Progressive Web App (PWA) designed to convert dense academic research and complex textbook excerpts into concise 2-sentence summaries and relatable real-world mental models for non-native English speakers.

Language Barriers Lock Millions Out of Learning



The Barrier

Academic literature lives strictly in complex, administrative English. ESL middle and high school students struggle not due to a lack of capability, but because vocabulary acts as a wall.

The Structural Gap

Standard translators perform word-for-word translation, which destroys scientific context and fails to explain the underlying concepts.

The Solution Fit

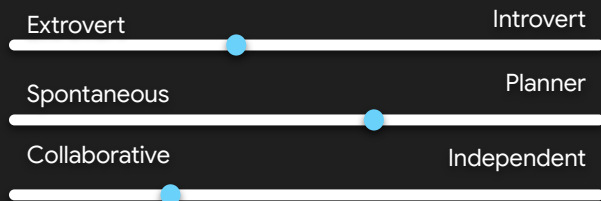
Students need tailored explanations using simplified language and real-world analogies rather than plain dictionary definitions.

Hamzah (Grade 8 Student)

BEHAVIORS

- Studies science and history using standard Grade 8 school textbooks
- Often searches online or uses basic translation tools when stuck on hard words
- Prefers short, simple explanations over long dictionary definitions

PERSONALITY



CURRENT FEELINGS

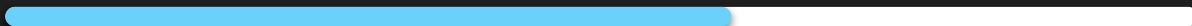


CHALLENGES

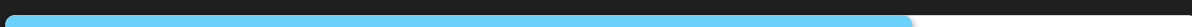
- Dense administrative and academic vocabulary blocks him from understanding lessons
- Standard translation tools translate word-for-word, destroying scientific context
- Spends extra time decoding complex English instead of learning the core concept

MOTIVATIONS

Understand school topics easily and perform well in class



Score higher on exams and keep up easily with the rest of his class



Study independently without needing teachers or parents to constantly translate textbook jargon



Core Mechanics

The Solution: How LangSlayer works

2-Sentence Summaries

Strips away dense administrative vocabulary to deliver the core scientific mechanism in two plain-English sentences.

Real-World Analogies

Pairs theoretical concepts with relatable everyday mental models (like explaining quantum superposition using a spinning coin).

Sub-Second Speed

Powered by Groq API running open-source models ([openai/gpt-oss-20b](#)) for instant, zero-wait simplification .

Built for Speed, Scale, and Reliability

- Frontend: Vite, Vanilla JS, and Tailwind CSS using a custom editorial grid-paper design (#F7F6F0 background and #101E36 text).
- AI Inference: Groq API running openai/gpt-oss-20b for sub-second responses, parsed with marked.js.
- PWA & Offline Mode: Installable progressive web app featuring client-side key storage and preset mock responses for zero-setup evaluation.

Disclosures & Future Roadmap

Q3 2026

Phase 1 (Q3 2026): Hackathon Build & Disclosures

- AI Model: Groq API running [openai/gpt-oss-20b](#) for open-source inference
- AI Tools: Gemini AI used for UI scaffolding and CSS refining.

Q4 2026

Phase 2 (Q4 2026): Interactive Glossaries

- Hover definitions for complex technical jargon directly within text excerpts.
- Instant inline audio pronunciation for non-native speakers.

Q1 2027

Phase 3 (Q1 2027): Regional Language Translation

- Side-by-side translation into regional languages (e.g., Hindi, Tamil, Telugu).
- Preserves scientific context while displaying local scripts.

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Thank you!

Try it out for yourself ☐

LangSlayer